

HARDENING VALIDATION REPORT

Phase 10 | Windows, SMB, Logging, and Attack Validation

Report ID	SOC-HL-P10-2026-0711	Result	PASSED
Date	11 July 2026	Status	Closed — Lab Exercise
Target	WIN11-01 / 192.168.10.20	Source	KALI-01 / 192.168.10.30
Scope	Hardening and validation	Primary control	Account lockout

Executive Summary

Phase 10 strengthened the SOC lab with account-lockout controls, service and protocol verification, host firewall enforcement, reduced remote-access exposure, and renewed logging checks. A final NetExec validation against the local Administrator account returned STATUS_ACCOUNT_LOCKED_OUT. Splunk also recorded post-hardening failed logons and an account-lockout event, while no new successful localadmin network logon appeared after the hardening window. The evidence confirms that the lockout and monitoring controls operated as intended.

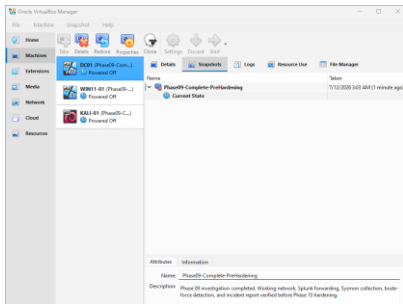
OUTCOME	The final validation did not obtain authenticated access. Windows enforced account lockout, and Splunk preserved evidence of the defensive response.
---------	--

Before and After Summary

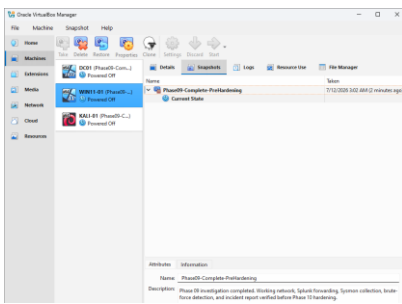
Control	Before	After	Assessment
Domain lockout	Threshold 0	5 attempts / 15 min	Improved
Local lockout	Never locked	5 attempts / 15 min	Improved
SMB1	Disabled	Disabled	Already compliant
SMB signing	Required	Required	Verified
RDP	Not baselined	Disabled	Reduced exposure
Firewall	Not baselined	All profiles ON	Verified
Logging	Operational	Operational	Revalidated
Attack outcome	Phase 9 access succeeded	Account locked	Blocked

Recovery Point and Pre-Hardening Baseline

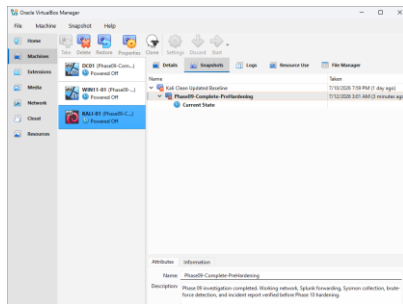
Before applying controls, the lab state was preserved for all three virtual machines. The baseline captures the domain policy, local account policy, and SMB configuration used for the final comparison.



DC01 recovery snapshot



WIN11-01 recovery snapshot



KALI-01 recovery snapshot

Baseline Findings

Area	Setting	Before value	Interpretation
Domain	Complexity	True	Already enabled
Domain	Min length	7	Retained
Domain	Password history	24	Retained
Domain	Lockout threshold	0	No domain lockout
Local	Min length/history	7 / 20	Retained
Local	Lockout threshold	Never	No local lockout
SMB	SMB1 / SMB2	False / True	SMB1 already disabled
SMB	Signing required	True	Already required
BASELINE NOTE		SMB1 and mandatory server signing were already configured securely before Phase 10; they were verified rather than newly enabled.	

Baseline Evidence

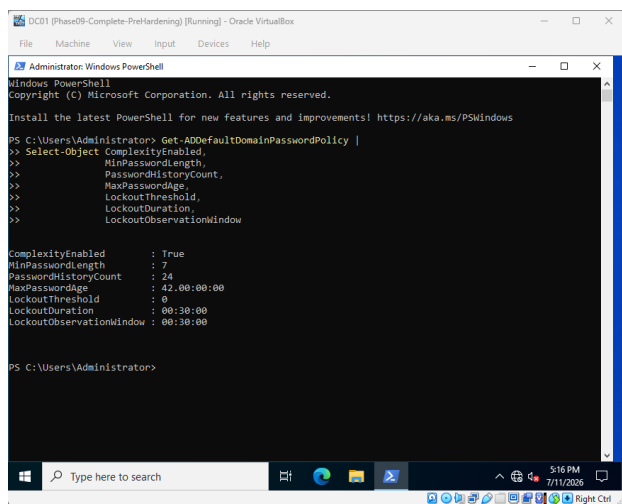


Figure 1. Domain policy before hardening.

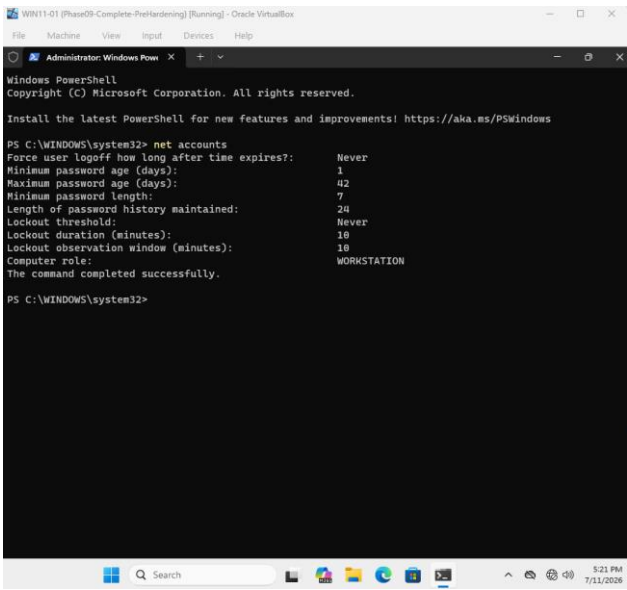


Figure 2. Local account policy before hardening.

SMB Baseline

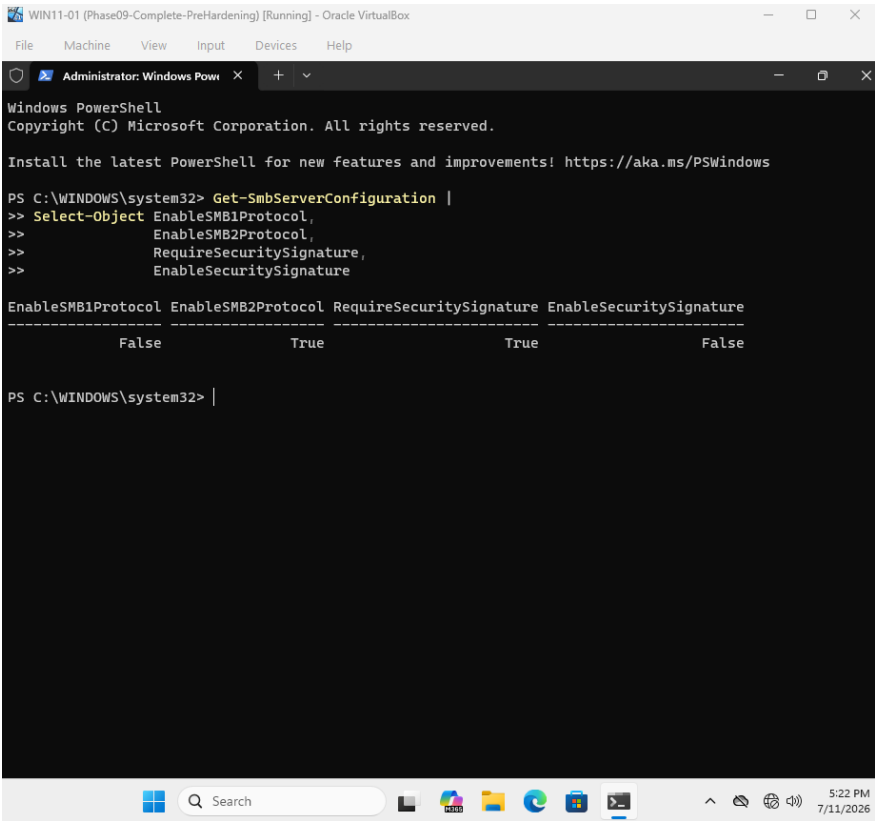


Figure 3. SMB1 was disabled and server signing was already required before Phase 10.

Password and Account-Lockout Controls

The principal security change was enabling account lockout for both the domain and the Windows 11 local account policy. Password complexity, minimum length, and history settings remained in place.

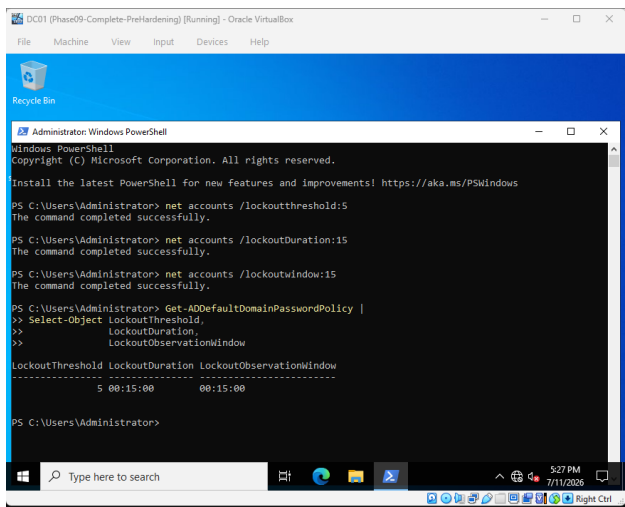


Figure 4. Domain lockout policy after hardening.

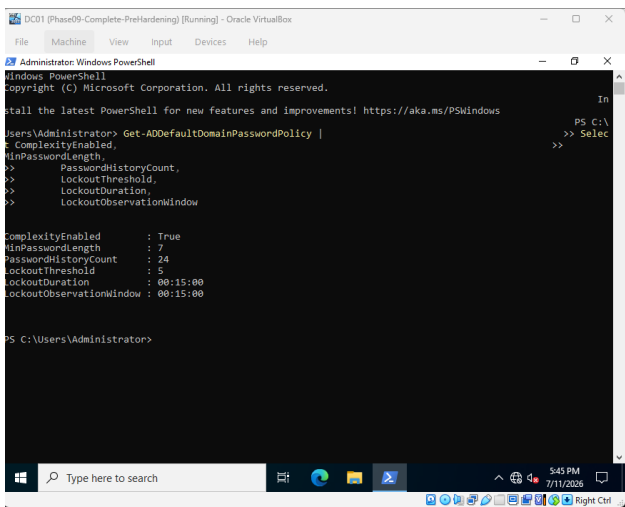


Figure 5. Domain password policy after hardening.

Local Lockout Policy

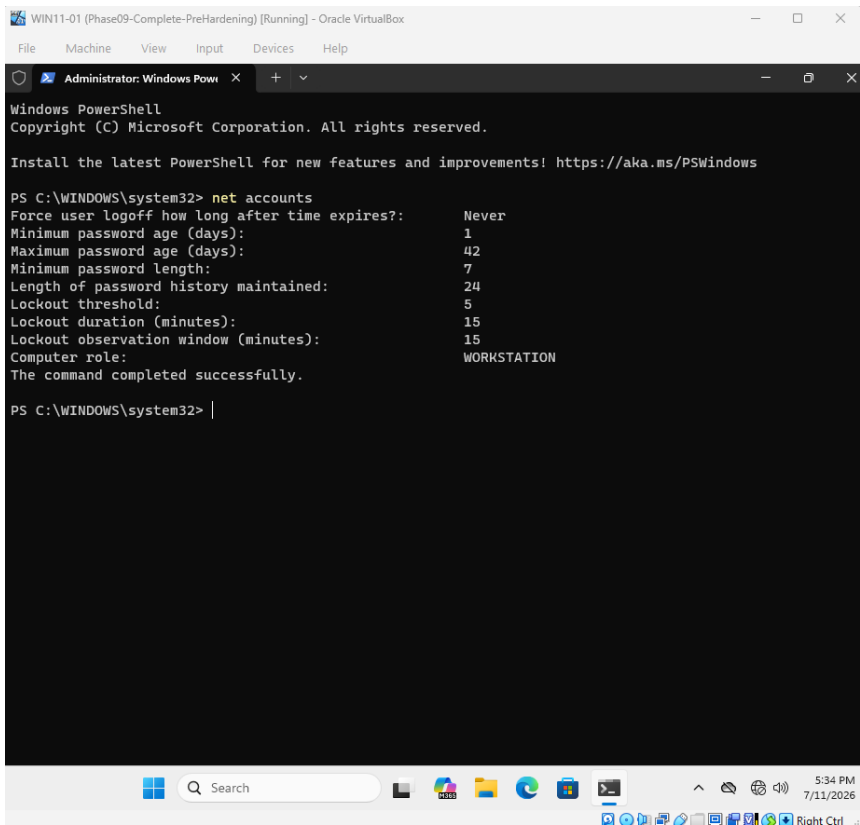


Figure 6. WIN11-01 local accounts lock after five failures for 15 minutes.

Account and SMB Hardening Verification

Local Account State

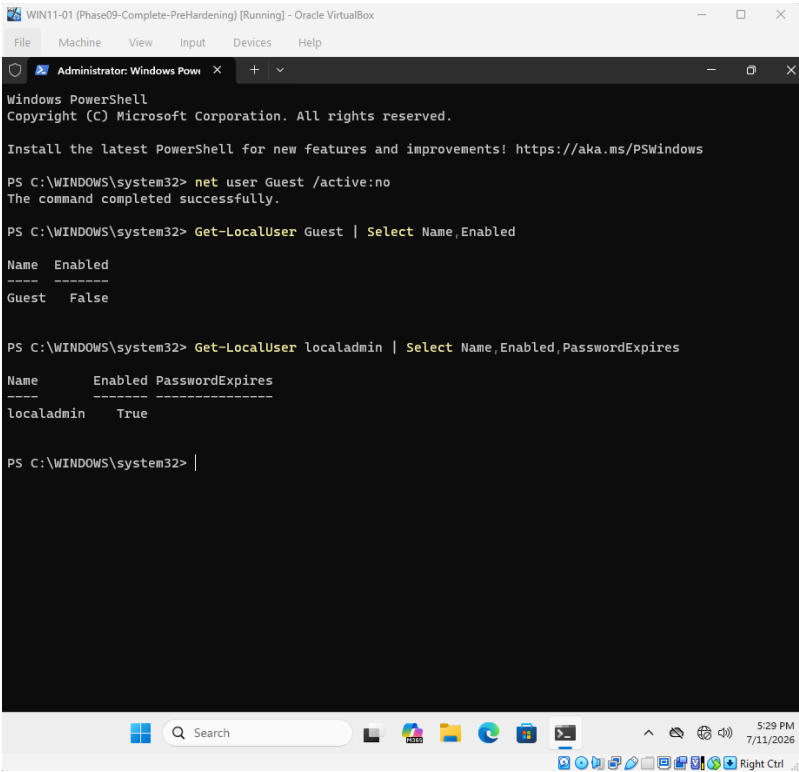


Figure 7. Guest is disabled; localadmin remained enabled for the controlled test.

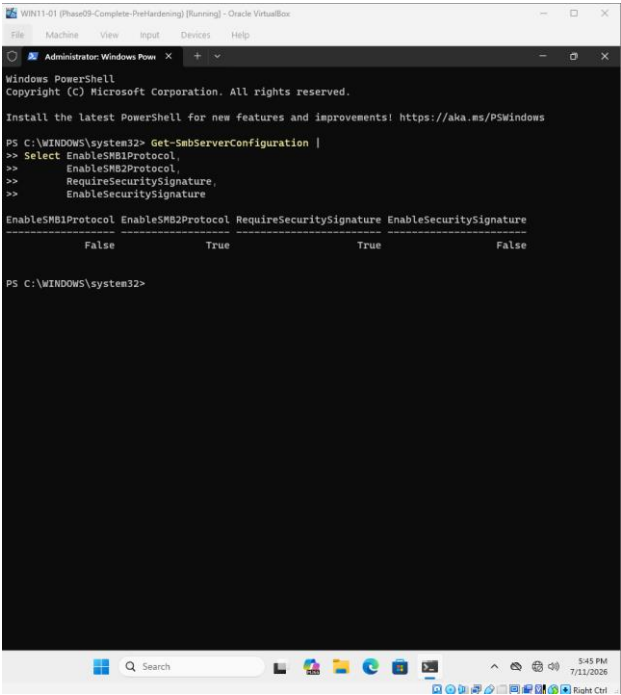


Figure 8. SMB configuration verified.

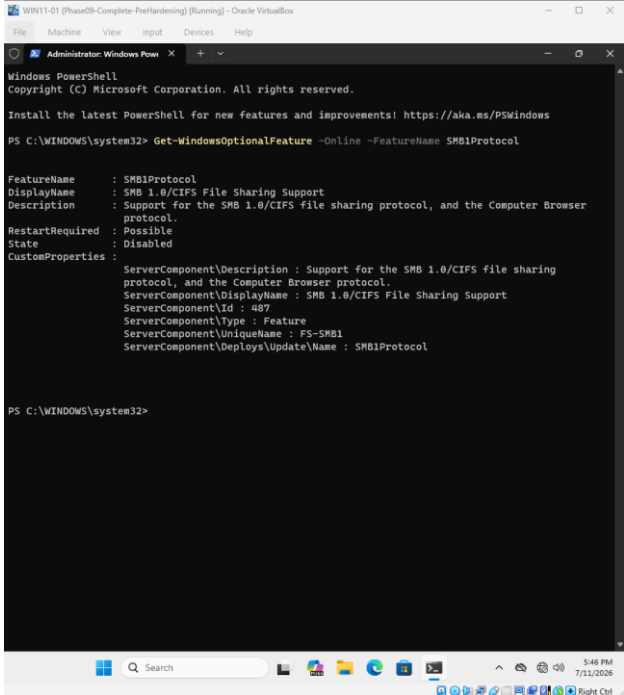


Figure 9. SMB1 optional feature disabled.

ASSESSMENT

The SMB evidence confirms secure settings but does not show a before/after change: SMB1 and required signing were already present in the baseline.

Remote Access and Host Firewall

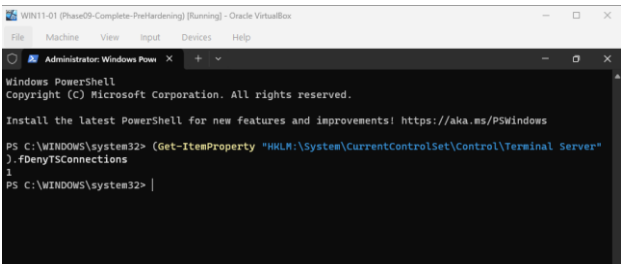


Figure 10. RDP disabled.

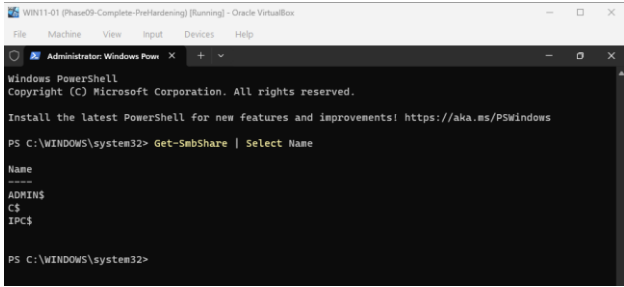


Figure 12. Default administrative shares remain.

Windows Firewall

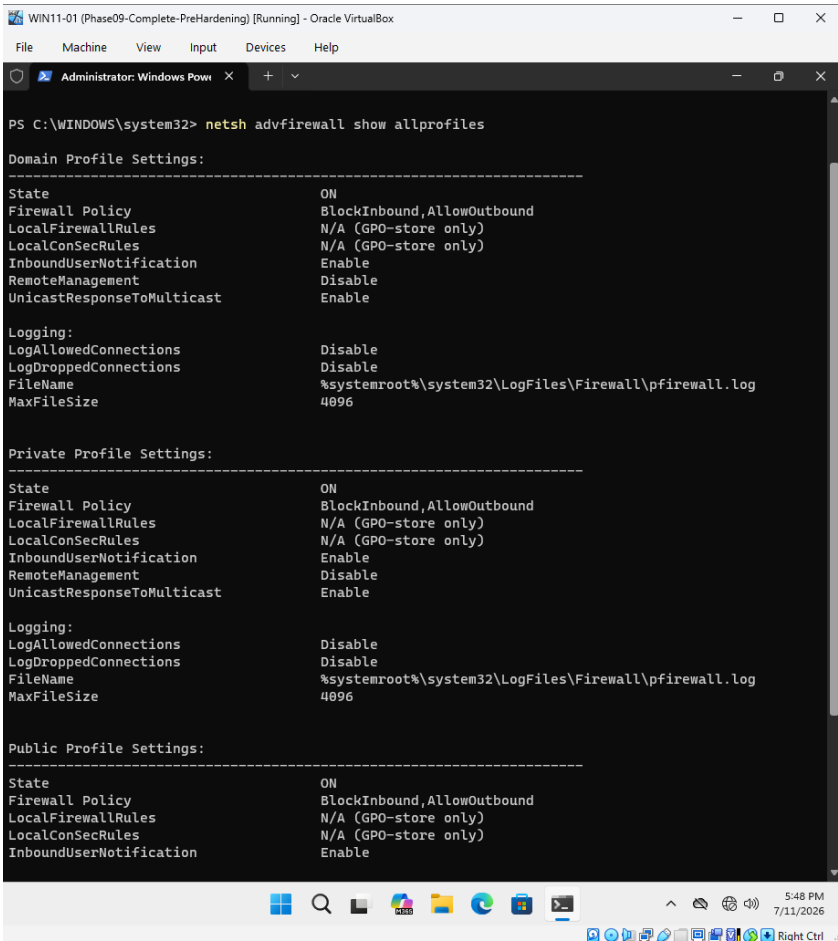


Figure 11. All Windows Firewall profiles are enabled with inbound traffic blocked by default.

RESIDUAL RISK

ADMIN\$, C\$, and IPC\$ remain available. In a production environment, access should be limited to approved management hosts through firewall and administrative controls.

Logging and Telemetry Health

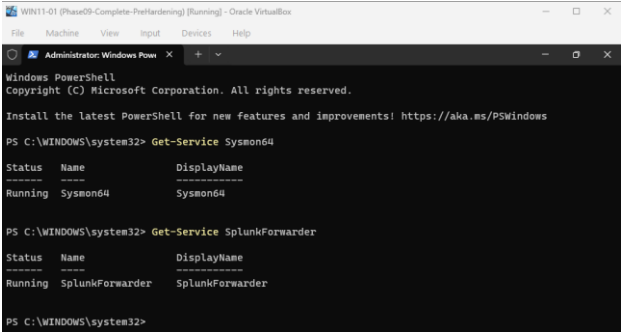


Figure 13. Sysmon and SplunkForwarder services running.

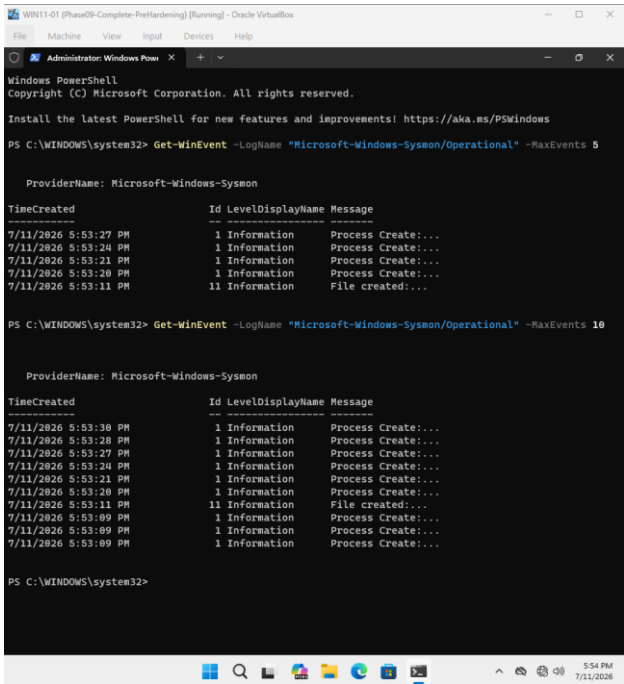


Figure 14. Live Sysmon events on WIN11-01.

Security Log Ingestion

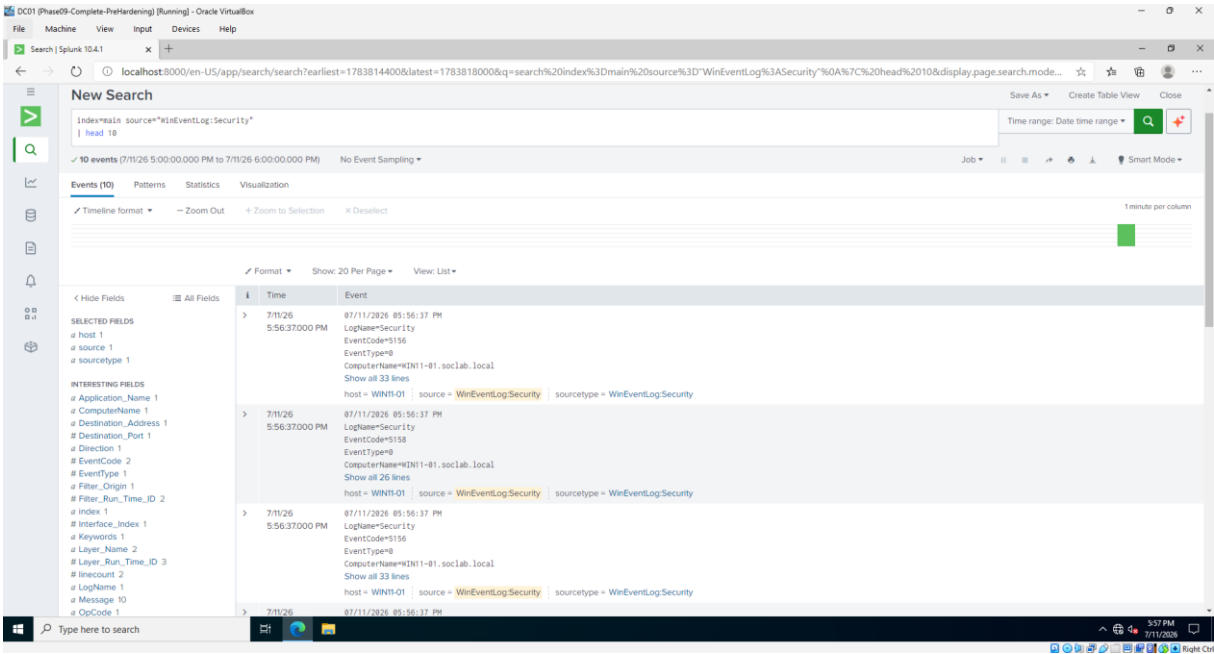


Figure 15. Windows Security logs continue to arrive in Splunk after hardening.

SIEM Detection Readiness

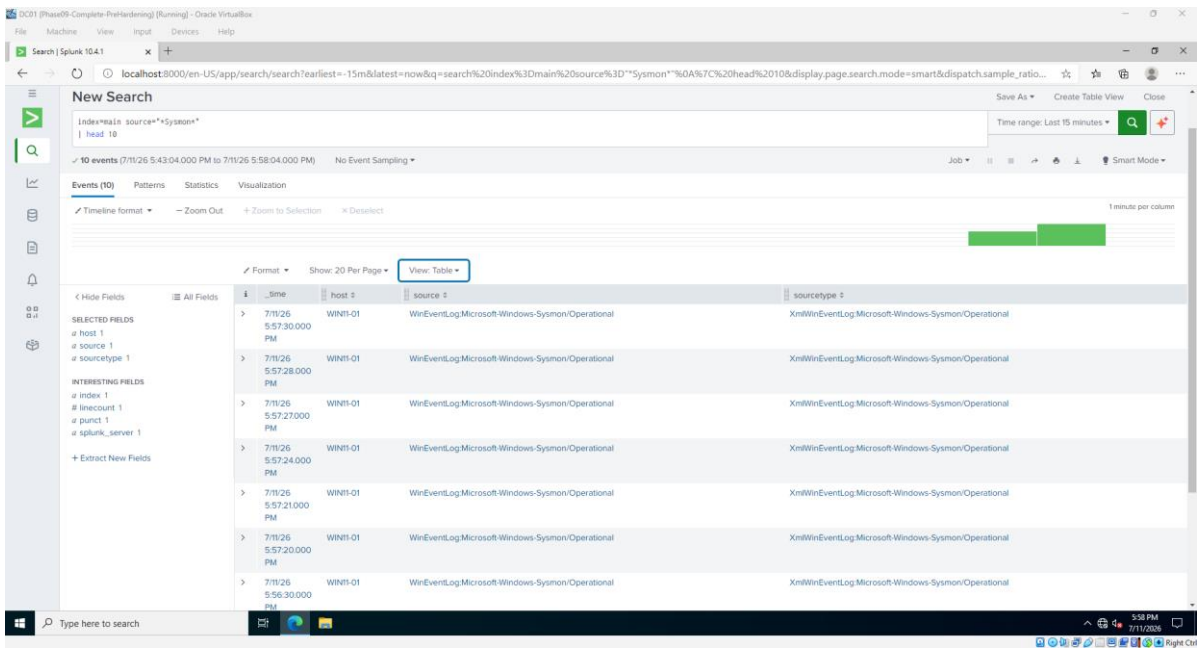


Figure 16. Sysmon logs are received and searchable in Splunk.

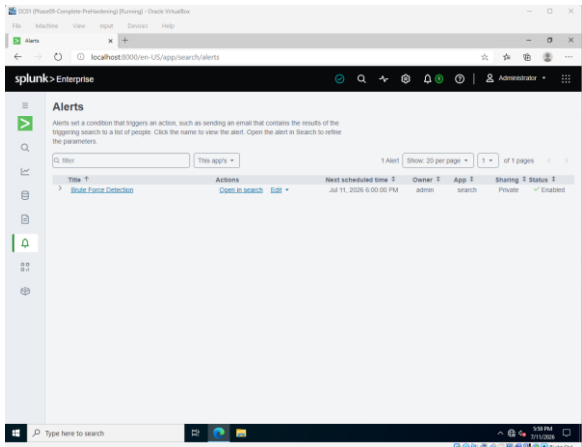


Figure 17. Brute Force Detection alert enabled.

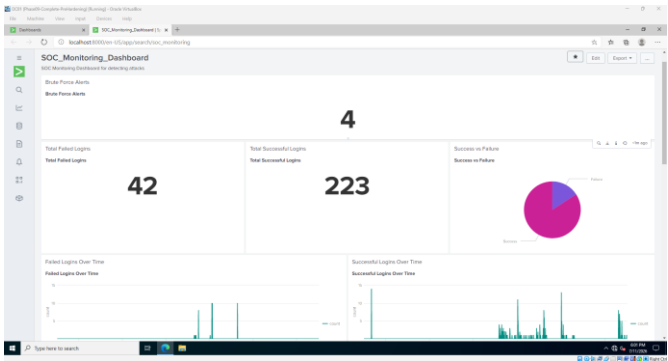


Figure 18. SOC monitoring dashboard healthy.

VALIDATION

The log pipeline, scheduled alert, and dashboard remained operational after the host and account-policy changes.

Final NetExec Validation

KALI-01 repeated an authorized SMB authentication test against WIN11-01 using local authentication. NetExec identified Windows 11 with SMB signing enabled and returned STATUS_ACCOUNT_LOCKED_OUT for the local Administrator account. Several SMB/NetBIOS timeout messages were also present, but no successful authentication marker appeared.

```
(kali㉿kali)-[~]
$ nxc smb 192.168.10.20 -u Administrator -p passwords.txt --local-auth
SMB 192.168.10.20 445 WIN11-01 [*] Windows 11 / Server 2025 Build 26100 x64 (name:WIN11-01) (d
omain:WIN11-01) (signing:True) (SMBv1:None)
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
SMB 192.168.10.20 445 WIN11-01 [-] WIN11-01\Administrator:0x0504557660x0 STATUS_ACCOUNT_LOCKED
_OUT
SMB 192.168.10.20 445 WIN11-01 [-] Connection Error: The NETBIOS connection with the remote ho
st timed out.
```

Figure 19. NetExec receives STATUS_ACCOUNT_LOCKED_OUT; no successful authentication is shown.

RESULT

PASSED — the lockout control prevented authenticated access during the final validation.

Validation Scope

In Phase 9, testing focused on localadmin. The final NetExec check used the local Administrator account with --local-auth. This confirms that the local account-lockout policy is enforced. The Splunk evidence below separately documents post-hardening failures for localadmin and confirms that no later successful localadmin network logon was recorded.

Validation point	Before hardening	After hardening
Authentication outcome	Four successful localadmin logons	Administrator locked; no NetExec success
Lockout event	Not enforced	Event ID 4740 recorded
Failed logons	Detected	Detected after hardening
Monitoring	Operational	Remained operational

Splunk Validation After Hardening

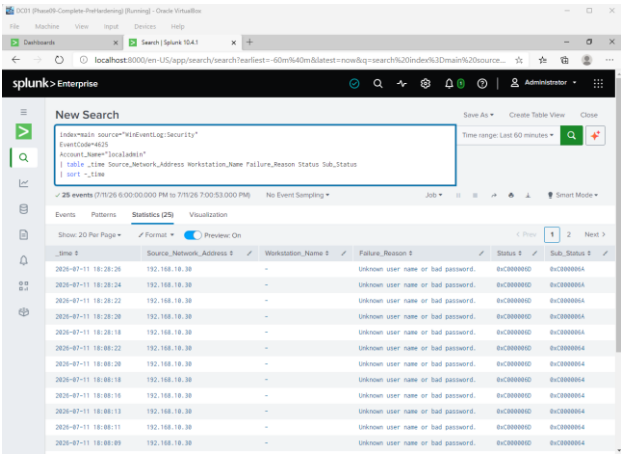


Figure 20. Failed localadmin logons after hardening.

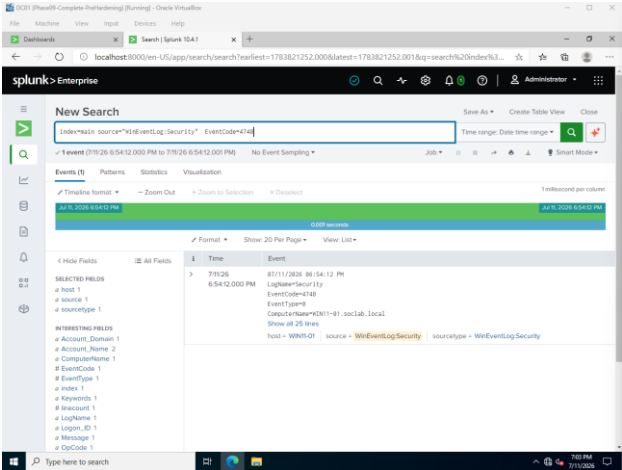


Figure 21. Account-lockout event recorded.

Successful Logon Review

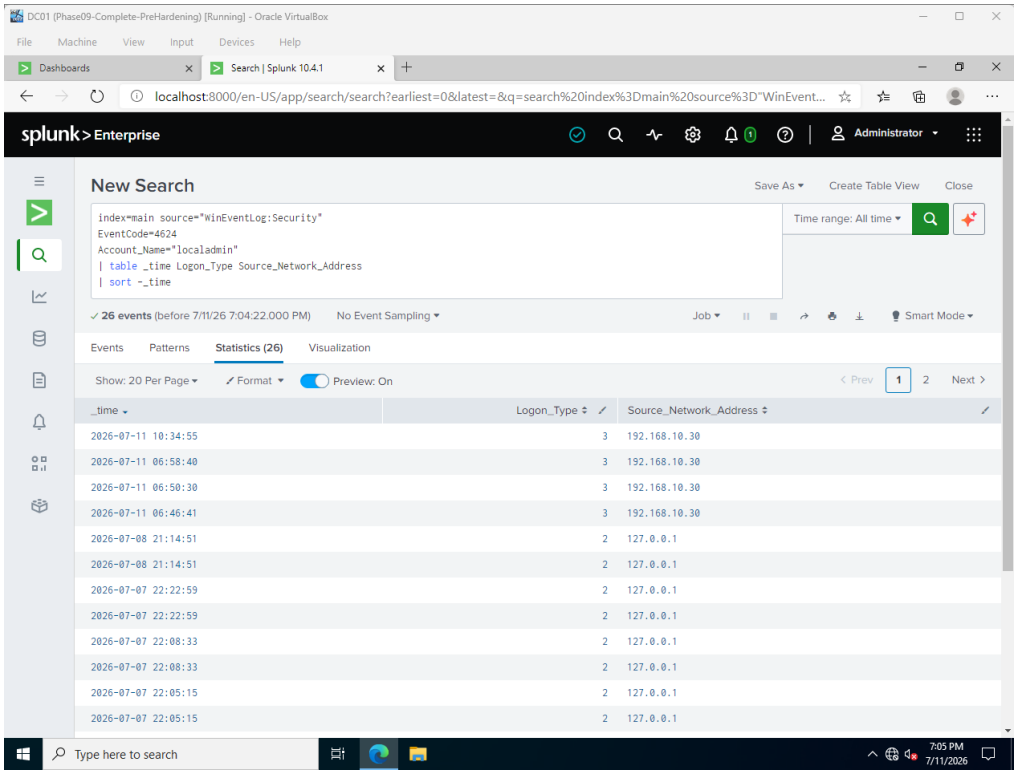


Figure 22. No new localadmin Event ID 4624 appears after hardening; the newest listed success remains 10:34:55.

EVIDENCE NOTE

The screenshots document separate validation steps. The 4625 search covers localadmin failures, the NetExec test covers Administrator, and Event ID 4740 confirms that WIN11-01 enforced account lockout. Together they show the control working across the host without presenting them as one continuous command sequence.

Security Assessment and Recommendations

Controls Demonstrated

- Domain and local account-lockout thresholds set to five attempts with 15-minute duration and observation windows.
- SMB1 disabled and SMB server signing required.
- RDP disabled and all Windows Firewall profiles enabled with inbound traffic blocked by default.
- Sysmon and Splunk Universal Forwarder services healthy, with Security and Sysmon data visible in Splunk.
- Brute-force alert enabled and SOC dashboard operational.
- NetExec received STATUS_ACCOUNT_LOCKED_OUT and no successful validation authentication was observed.

Residual Risks and Limitations

- Default administrative shares ADMIN\$, C\$, and IPC\$ remain enabled.
- The final NetExec validation used Administrator, while the Phase 9 baseline used localadmin.
- The 4740 event predates the final NetExec screenshot and may reflect the earlier manual lockout validation.
- The no-success search uses an all-time window; the conclusion depends on the newest displayed 4624 timestamp being earlier than the hardening validation.
- Password minimum length remained seven characters and should be increased for a production environment.

Recommended Next Improvements

- Increase password minimum length to at least 14 characters and use unique service/admin credentials.
- Limit SMB and administrative-share access to approved management hosts or a dedicated administration subnet.
- Enable Windows Firewall dropped-packet logging and forward those logs to the SIEM.
- Repeat the final validation with localadmin for a strict same-account before/after comparison.
- Tune the Splunk alert to correlate failures, lockout, and any later success by host, source IP, and account.

Final Conclusion

Phase 10 is complete. The lab preserved a recoverable pre-hardening state, introduced effective lockout controls, verified protocol and host protections, confirmed telemetry health, and demonstrated that the final local-authentication test did not obtain access. The project now covers the complete defensive cycle: build, attack, detect, investigate, harden, and validate.

PROJECT STATUS	SOC Home Lab technical objectives completed. The final evidence set is suitable for academic or portfolio use, with the validation scope clearly documented.
----------------	--

Evidence Register

The filenames below map the submitted evidence to the control or validation point documented in this report. Numbering follows the original evidence package.

ID	Evidence file	Purpose
01	DC01 / WIN11 / KALI snapshots	Recovery points
02	Domain-Policy-Before-Hardening	Domain baseline
03	Local-Account-Policy-Before	Local baseline
04	SMB-Configuration-Before	SMB baseline
05	Domain-Lockout-Policy-After	Domain lockout
06	LocalAdmin-Status	Account state
07	Local-Lockout-Policy-After	Local lockout
08	Domain-Password-Policy-After	Domain verification
09	SMB-Hardening	SMB verification
10	SMB1-Disabled	Legacy protocol disabled
11	RDP-Disabled	Remote-access reduction
12	Firewall-Enabled	Host firewall
13	Administrative-Shares	Residual exposure
14	Sysmon & Forwarder Services	Service health
16	Sysmon-Live-Events	Local telemetry
17	Security-Logs-Received	Security ingestion
18	Sysmon-Logs-Received	Sysmon ingestion
19	BruteForce-Alert-Enabled	Detection readiness
20	SOC-Dashboard-Healthy	Monitoring health
21	NetExec-Blocked	Final attack validation
23	4625-After-Hardening	Failed logons
24	Account-Lockout-Event	Event ID 4740
25	No-Successful-Logon	No later localadmin 4624